

FastAPI Notes for RAG Testing

1. What is FastAPI?

FastAPI is a modern, high-performance Python web framework used for building APIs. It is based on standard Python type hints and is known for speed, automatic validation, and interactive API documentation.

2. Key Features

Fast performance, automatic OpenAPI documentation, request validation using Pydantic, async support, dependency injection, and easy integration with databases.

3. Installation

Install FastAPI using `pip install fastapi`. For running applications, `uvicorn` is commonly used.

4. Basic Example

from fastapi import FastAPI. Create `app = FastAPI()` and define routes using decorators such as `@app.get('/')`.

5. Path Parameters

Path parameters allow dynamic URLs. Example: `/users/{user_id}`. FastAPI automatically converts data types based on type hints.

6. Query Parameters

Query parameters are values passed after a question mark in the URL, such as `/items?limit=10`.

7. Request Body

FastAPI uses Pydantic models to validate and parse request bodies automatically.

8. Response Models

Response models help control output data and generate accurate API documentation.

9. Dependency Injection

Dependencies are reusable components that can be injected into routes using `Depends()`.

10. Async Support

FastAPI supports asynchronous programming using `async def`, making it suitable for high-concurrency applications.

FastAPI Interview Notes

Q: Why is FastAPI fast?

A: It uses Starlette for web functionality and Pydantic for data validation.

Q: What server is commonly used with FastAPI?

A: Uvicorn.

Q: What is Pydantic?

A: A library used for data validation and serialization using Python type hints.